OFFICIAL TECHNICAL SPECIFICATION DATASHEET
MANUFACTURER: LONGI SOLAR
MODEL / SERIES: Hi-MO 7 LR5-72HGD-580M
DOCUMENT TITLE: LONGi Solar Hi-MO 7 LR5-72HGD Dual-Glass Bifacial Module Technical Datasheet
REVISION / DATE: LR5-72HGD-202305-V01 | Published: 2023-05-18

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1. PRODUCT IDENTIFICATION & ARCHITECTURE

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|---|---|
| Cell Technology: | N-type TOPCon |
| Module Construction: | Bifacial (Glass-Glass) |
| Number of Cells / Format: | 144 (6x24) N-type Bifacial Half-cut Cells |
| Glass / Front Encapsulant: | Dual Glass 2.0 + 2.0 mm Semi-tempered Glass |
| Frame Material: | Anodized Aluminium Alloy |
| Junction Box / Protection: | IP68 with 3 bypass diodes |

2. ELECTRICAL CHARACTERISTICS AT STC (1000 W/m2, 25C, AM 1.5)

| | |
|---|---|
| Nominal Maximum Power (Pmax / Pnom): | 580.0 Wp |
| Module Aperture Efficiency: | 22.50 % |
| Open Circuit Voltage (Voc): | 51.41 V |
| Short Circuit Current (Isc): | 14.22 A |
| Maximum Power Voltage (Vmp): | 43.22 V |
| Maximum Power Current (Imp): | 13.42 A |
| Maximum System Voltage: | 1500.0 V DC |
| Maximum Series Fuse Rating: | 30.0 A |

3. TEMPERATURE CHARACTERISTICS & OPERATING LIMITS

| | |
|---|---|
| Temp. Coefficient of Pmax (gamma): | -0.280 %/C |
| Temp. Coefficient of Voc (beta): | -0.230 %/C |
| Temp. Coefficient of Isc (alpha): | +0.045 %/C |
| Nominal Cell Temperature (NOCT/NMOT): | 45 +/- 2 C (NOCT) |
| Operating Temperature Range: | -40 C to +85 C |

4. MECHANICAL DIMENSIONS & RATINGS

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|---|---|
| Length: | 2278.0 mm |
| Width: | 1134.0 mm |
| Thickness (Height): | 30.0 mm |
| Gross Aperture Area: | 2.5833 m2 |
| Module Weight: | 31.8 kg |
| Mechanical Design Load (Front): | 5400 Pa (Snow load) |
| Mechanical Design Load (Rear): | 2400 Pa (Wind load) |

5. WARRANTY & LIFETIME PERFORMANCE

| | |
|---|---|
| Product Workmanship Warranty: | 12.0 Years |
| Linear Power Output Warranty: | 30.0 Years |
| Annual Linear Degradation: | 0.40 % / year |
| End of Warranty Retained Output: | 88.2 % |

6. PROVENANCE & ARCHIVAL METADATA

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SolarSelect MLOps Hardware Database Reference Standard IEC 61215 / IEC 61730